

Skylark Drones — BI Agent Decision Log

Technical Assignment: Monday.com Business Intelligence Agent | Hosted prototype: skylark-bi-agent-six-beta.vercel.app

1. Architecture Decision

Chosen architecture: A Next.js application with a server-side API route, Gemini for natural-language query understanding and tool selection, Monday.com API for live read-only data, and dedicated normalization/analytics modules.

The flow is **Founder question** → **Gemini** → **business-tool selection** → **Monday.com retrieval** → **normalization** → **deterministic analytics** → **executive response**. This keeps numerical business logic outside the language model and reduces the risk of invented figures.

2. Technology Choices

Component	Decision & rationale
Next.js	Used for the conversational UI and server-side API endpoints. It keeps the prototype simple to develop and deploy while keeping credentials off the client.
Gemini	Used for natural-language understanding, business-tool selection, clarification, and concise founder-level explanations. Business numbers come from deterministic tools.
Monday.com API	Chosen over MCP because the assignment requires read-only access to two boards and the API gives a direct, predictable integration for this scope.
TypeScript	Used for type safety across the application and to catch integration/schema errors during development.

3. Data Resilience & Normalization

The source boards contain real-world inconsistencies. Raw Monday.com records are normalized before analysis: dates are converted into consistent representations, numeric/value fields are parsed safely, and text/sector labels are normalized. Missing values remain missing rather than silently becoming zero. The agent reports important data-quality limitations when they affect interpretation or forecasting.

4. Query Understanding & Routing

Gemini receives three narrowly defined business functions. Pipeline, deal, revenue and sales-stage questions route to sales analysis; work execution, billing, collections and receivables route to operations analysis; questions spanning both route to cross-board analysis. A sector is passed through only when the user specifies one. Genuinely ambiguous requests are intended to receive a concise clarification rather than a guess.

5. Cross-Board Analysis

Cross-board questions load both live boards, apply the same normalization rules, and pass both datasets to a dedicated cross-board analytics function. This supports comparisons such as a sector's sales pipeline versus project execution, billing and receivables.

6. Error Handling & Security

Monday.com and AI calls are protected with error handling. Tool failures are converted into controlled responses rather than exposing implementation details. API keys are stored in environment variables, are not committed to Git, and are not exposed to the browser.

7. Key Assumptions

- The two Monday.com boards are the authoritative sources for the assignment.
- The agent is read-only and does not modify Monday.com records.
- Missing data should remain explicitly missing and should not be inferred.
- Sector refers to the normalized category represented in the board data.
- Live Monday.com data is preferred over hardcoded sample CSV values.

8. Trade-offs

The implementation prioritizes correctness, transparency and completion within the six-hour constraint over advanced forecasting or a larger orchestration framework. The API-based approach is simpler than adding MCP infrastructure, while dedicated analytics functions make numerical outputs more deterministic. The trade-off is that the prototype does not attempt enterprise authentication, long-term memory, or advanced predictive modeling.

9. Leadership Updates Interpretation

The optional requirement was interpreted as helping prepare leadership-ready updates from sales and operational data. The agent therefore emphasizes a headline, key numbers, interpretation, data-quality caveats, and relevant risks/opportunities. This turns raw board data into concise executive context that can be reused in leadership updates.

10. What I Would Improve With More Time

I would add configurable time-period filters such as quarter-to-date, trend visualizations, automated leadership-summary generation, stronger schema validation, caching/rate-limit handling for Monday.com, and broader automated tests for messy records. Before production use, I would also add authentication and role-based access controls.

Submission note: The prototype is hosted on Vercel and the source code is maintained in GitHub. Environment secrets are intentionally excluded from source control.